

Vydhra Machine Learning Job-Ready Bootcamp

Duration: 8 Weeks	Format: 2 Topic Points + Practice	Level: Beginner to Job-Ready Foundation
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Website-Style Course Structure

Module 01	Week 1 - Machine Learning Foundations		
Topics Covered	Introduction to machine learning, real-world use cases, supervised vs unsupervised learning		
	ML workflow, Python environment setup, and essential tools: NumPy, Pandas, Scikit-learn		
Practice	Dataset exploration and basic ML workflow implementation		
Outcome	Students understand machine learning fundamentals and can set up their environment for practical work.		

Module 02	Week 2 - Data Preprocessing & Feature Engineering		
Topics Covered	Data cleaning, handling missing values, encoding categorical variables, and feature scaling		
	Train-test split, feature engineering basics, and understanding data leakage		
Practice	Real-world dataset cleaning and preprocessing project		
Outcome	Students can prepare raw data and transform it into a format ready for machine learning models.		

Module 03	Week 3 - Regression Models		
Topics Covered	Linear regression, cost function, model fitting, and evaluation using RMSE and R-squared		
	Overfitting vs underfitting, validation basics, and interpreting regression results		
Practice	House price prediction project		
Outcome	Students can build, evaluate, and interpret regression models with confidence.		

Module 04	Week 4 - Classification Techniques	
Topics Covered	Logistic regression, Decision Tree, KNN, and Naive Bayes for classification problems	
	Confusion matrix, accuracy, precision, recall, and F1-score for model evaluation	
Practice	Customer churn prediction project	
Outcome	Students can solve classification problems and assess model performance effectively.	

Module 05	Week 5 - Model Optimization & Advanced Algorithms	■
Topics Covered	Random Forest, Gradient Boosting, model comparison, and algorithm selection	
	Cross-validation, hyperparameter tuning, and improving model performance	
Practice	Model performance improvement and comparison task	
Outcome	Students can optimize models and identify the best-performing approach for a dataset.	

Module 06	Week 6 - Unsupervised Learning & Pattern Discovery	■
Topics Covered	K-Means clustering, practical clustering use cases, and customer segmentation workflows	
	Principal Component Analysis (PCA) and interpreting hidden patterns in data	
Practice	Customer segmentation project	
Outcome	Students can identify patterns, group data, and explain clustering results in practical scenarios.	

Module 07	Week 7 - Deep Learning Fundamentals	■
Topics Covered	Neural networks, layers, activation functions, and the core ideas behind deep learning	
	TensorFlow/Keras introduction and when to use deep learning vs traditional machine learning	
Practice	Basic image classification demonstration	
Outcome	Students gain a strong beginner-level understanding of deep learning concepts and tools.	

Module 08	Week 8 - Capstone & Job Readiness	■
Topics Covered	End-to-end machine learning project workflow, model evaluation, interpretation, and deployment basics	
	GitHub portfolio polish, project presentation, and machine learning interview preparation	
Practice	Final capstone project using a real-world dataset	
Outcome	Students complete a portfolio-ready machine learning project and strengthen entry-level job readiness.	